INCH-POUND

MIL-M-38510/327B
15 September 2003
SUPERSEDING
MIL-M-38510/327A
10 December 1982

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, LOW-POWER SCHOTTKY TTL, COUNTERS, MONOLITHIC SILICON

Inactive for new design after 18 April 1997.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, low-power Schottky TTL, binary and decade counters microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided for each type and are reflected in the complete part number. For this product, the requirements of MIL-M-38510 have been superseded by MIL-PRF-38535, (see 6.3).

1.2 Part number. The part number should be in accordance with MIL-PRF-38535, and as specified herein.

1.2.1 Device types. The device types should be as follows:

Device type	Circuit
01	Dual decade counter with A and B inputs
02	Dual 4 bit binary counter
03	Dual decade counter with clear and set-to-nine

1.2.2 Device class. The device class should be the product assurance level as defined in MIL-PRF-38535.

1.2.3 Case outlines. The case outlines should be as designated in MIL-STD-1835 and as follows:

Outline letter	Descriptive designator	Terminals	Package style
A	GDFP5-F14 or CDFP6-F14	14	Flat pack
B	GDFP4-14	14	Flat pack
C	GDIP1-T14 or CDIP2-T14	14	Dual-in-line
D	GDFP1-F14 or CDFP2-F14	14	Flat pack
E	GDIP1-T16 or CDIP2-T16	16	Dual-in-line
F	GDFP2-F16 or CDFP3-F16	16	Flat pack
2	CQCC1-N20	20	Square leadless chip carrier

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAS, P. O. Box 3990, Columbus, OH 43216-5000, by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V to 7.0 V
Input voltage range	-1.5 V dc at -18 mA to 5.5 V dc
Storage temperature range	-65° to +150°C
Maximum power dissipation (P_D) 1/	143 mW dc
Lead temperature (soldering, 10 seconds)	300°C
Thermal resistance, junction to case (θ_{JC}):	
Cases A, B, C, D, E, F, and 2	(See MIL-STD-1835)
Junction temperature (T_J) 2/	+175°C

1.4 Recommended operating conditions.

Supply voltage (V_{CC})	4.5 V minimum to 5.5 V maximum
Minimum high level input voltage (V_{IH})	2.0 V
Maximum low level input voltage (V_{IL})	0.7 V
Case operating temperature range (T_C)	-55°C to +125°C
Normalized fanout (each output)	10 maximum
Width of input count pulse, $t_{P(IN)}$:	
Input A, clock	20 ns minimum
Input B	25 ns minimum
Width of clear pulse	20 ns minimum
Input clock frequency (F_{MAX}):	
Input A (Types 01, 02, 03)	0 to 25 MHz
Input B (Type 01)	0 to 12.5 MHz
Setup time (Types 01 and 02)	
Clear inactive state setup time (time for clear ↓ to A or B input ↓)	35 ns minimum

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and Standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Departments of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-38535 - Integrated Circuits (Microcircuits) Manufacturing, General Specification for.

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-883 - Test Method Standard for Microelectronics.
MIL-STD-1835 - Interface Standard Electronic Component Case Outlines

(Unless otherwise indicated, copies of the above specifications and standards are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

_____

1/ Must withstand the added P_D due to short-circuit test (e.g., I_{OS}).
2/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions in accordance with MIL-PRF-38535.

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2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Microcircuits furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.3 and 6.4).

3.2 Item requirements. The individual item requirements shall be in accordance with MIL-PRF-38535 and as specified herein or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not affect the form, fit, or function as described herein.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-PRF-38535 and herein.

3.3.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.3.2 Logic diagrams. The logic diagrams shall be specified on figure 2.

3.3.3 Truth table. The truth table shall be as specified on figure 3.

3.3.4 Schematic circuits. The schematic circuits shall be maintained by the manufacturer and made available to the qualifying activity and the preparing activity upon request.

3.3.5 Case outlines. The case outlines shall be as specified in 1.2.3.

3.4 Lead material and finish. The lead material and finish shall be in accordance with MIL-PRF-38535 (see 6.6).

3.5 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.6 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.7 Marking. Marking shall be in accordance with MIL-PRF-38535.

3.8 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 12 (see MIL-PRF-38535, appendix A).

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4. VERIFICATION

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-PRF-38535 or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not effect the form, fit, or function as described herein.

4.2 Screening. Screening shall be in accordance with MIL-PRF-38535 and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. The burn-in test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in test method 1015 of MIL-STD-883.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. Additional screening for space level product shall be as specified in MIL-PRF-38535, appendix B.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-38535.

4.4 Technology Conformance inspection (TCI). Technology conformance inspection shall be in accordance with MIL-PRF-38535 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table III of MIL-PRF-38535 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II MIL-PRF-38535.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table IV of MIL-PRF-38535 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. The steady-state life test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in test method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table V of MIL-PRF-38535. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified and as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$		Device type	Limits		Unit
					Min	Max	
High level output voltage	V_{OH}	V_{CC} = 4.5 V, V_{IH} = 2.0 V, V_{IL} = 0.7 V, I_{OH} = -400 µA		All	2.5		V
Low level output voltage	V_{OL}	V_{CC} = 4.5 V, V_{IL} = 0.7 V, V_{IH} = 2.0 V, I_{OL} = 4 mA		All		0.4	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -18 mA, T_C = +25°C		All		-1.5	V
High level input current at clear or set to 9 inputs	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.7 V		All		20	µA
High level input current at clear or set to 9 inputs	I_{IH4}	V_{CC} = 5.5 V, V_{IN} = 5.5 V		All		100	µA
High level input current at input A or clock	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 2.7 V		All		100	µA
High level input current at input A or clock	I_{IH5}	V_{CC} = 5.5 V, V_{IN} = 5.5 V		01, 03		200	µA
				02		400	
High level input current at input B	I_{IH3}	V_{CC} = 5.5 V, V_{IN} = 2.7 V		01		200	µA
High level input current at input B	I_{IH6}	V_{CC} = 5.5 V, V_{IN} = 5.5 V		01		400	µA
Low level input current at clear or set to 9 inputs	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.4 V		01, 02	-120	-400	µA
				03	-135	-400	
Low level input current at input A	I_{IL2}	V_{CC} = 5.5 V, V_{IN} = 0.4 V		All	-.35	-2.4	mA
Low level input current at input B	I_{IL3}	V_{CC} = 5.5 V, V_{IN} = 0.4 V		01	-.60	-3.2	mA
Short circuit output current	I_{OS}	V_{CC} = 5.5 V 1/		All	-15	-100	mA
Supply current	I_{CC}	V_{CC} = 5.5 V, V_{IN} = 0 V		All		26	mA
Maximum input clock frequency, A or CLK	F_{MAX1}	V_{CC} = 5.0 V, C_L = 50 pF, R_L = 2 kΩ		All	25		MHz
Maximum input clock frequency, B	F_{MAX2}	V_{CC} = 5.0 V, C_L = 50 pF, R_L = 2 kΩ		01	12.5		MHz
Propagation delay time, low-to-high	t_{PLH1}	V_{CC} = 5.0 V,	A to QC	01	2	84	ns
		C_L = 50 pF,	A to QA	02	2	33	
		R_L = 2 kΩ	CLK to QA	03	2	33	
Propagation delay time, high to low	t_{PHL1}	V_{CC} = 5.0 V,	A to QC	01	2	84	ns
		C_L = 50 pF,	A to QA	02	2	33	
		R_L = 2 kΩ	CLK to G	03	2	33	

1/ Not more than one output should be shorted at a time.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$		Device type	Limits		Unit
					Min	Max	
Propagation delay time, low-to-high	t_{PLH2}	V_{CC} = 5.0 V, C_L = 50 pF, R_L = 2 kΩ	B to QD	01	2	34	ns
			A to QD	02	2	93	
			CLK to QA	03	2	77	
Propagation delay time, high to low	t_{PHL2}	V_{CC} = 5.0 V, C_L = 50 pF, R_L = 2 kΩ	B to QD	01	2	34	ns
			A to QD	02	2	93	
			CLK to QC	03	2	77	
Propagation delay time, high-to-low level, CLR to Q	t_{PHL3}	V_{CC} = 5.0 V C_L = 50 pF, R_L = 2 kΩ		All	2	56	ns
Propagation delay time, low to high level, set to 9 to QA	t_{PLH4}	V_{CC} = 5.0 V C_L = 50 pF, R_L = 2 kΩ		03	2	57	ns
Propagation delay time, high to low level, set to 9 to QB	t_{PHL4}	V_{CC} = 5.0 V C_L = 50 pF, R_L = 2 kΩ		03	2	53	ns

TABLE II. Electrical test requirements.

MIL-PRF-38535 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters	1	1
Final electrical test parameters	1*, 2, 3, 7, 9, 10, 11	1*, 2, 3, 7, 9
Group A test requirements	1, 2, 3, 7, 8, 9, 10, 11	1, 2, 3, 7, 9, 10, 11
Group B electrical test parameters when using the method 5005 QCI option	1, 2, 3, 7, 8 9, 10, 11	N/A
Group C end-point electrical parameters	1, 2, 3, 7, 8 9, 10, 11	1, 2, 3
Group D end-point electrical parameters	1, 2, 3	1, 2, 3

*PDA applies to subgroup 1.

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Terminal number	Terminal symbol					
	Device type 01		Device type 02		Device type 03	
	Case E, F	Case 2	Case A, B, C, D	Case 2	Case E, F	Case 2
1	1A	NC	1A	NC	1CLK	NC
2	1CLR	1A	1CLR	1A	1CLR	1CLK
3	1QA	1CLR	1QA	1CLR	1QA	1CLR
4	1B	1QA	1QB	1QA	1 SET T0 9	1QA
5	1QB	1B	1QC	NC	1QB	1 SET T0 9
6	1QC	NC	1QD	1QB	1QC	NC
7	1QD	1QB	GND	NC	1QD	1QB
8	GND	1QC	2QD	1QC	GND	1QC
9	2QD	1QD	2QC	1QD	2QD	1QD
10	2QC	GND	2QB	GND	2QC	GND
11	2QB	NC	2QA	NC	2QB	NC
12	2B	2QD	2CLR	2QD	2 SET TO 9	2QD
13	2QA	2QC	2A	2QC	2QA	2QC
14	2CLR	2QB	V_{CC}	2QB	2CLR	2QB
15	2A	2B		NC	2CLK	2 SET TO 9
16	V_{CC}	NC		2QA	V_{CC}	NC
17		2QA		NC		2QA
18		2CLR		2CLR		2CLR
19		2A		2A		2CLK
20		V_{CC}		V_{CC}		V_{CC}

FIGURE 1. Terminal connections.

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DEVICE TYPE 01

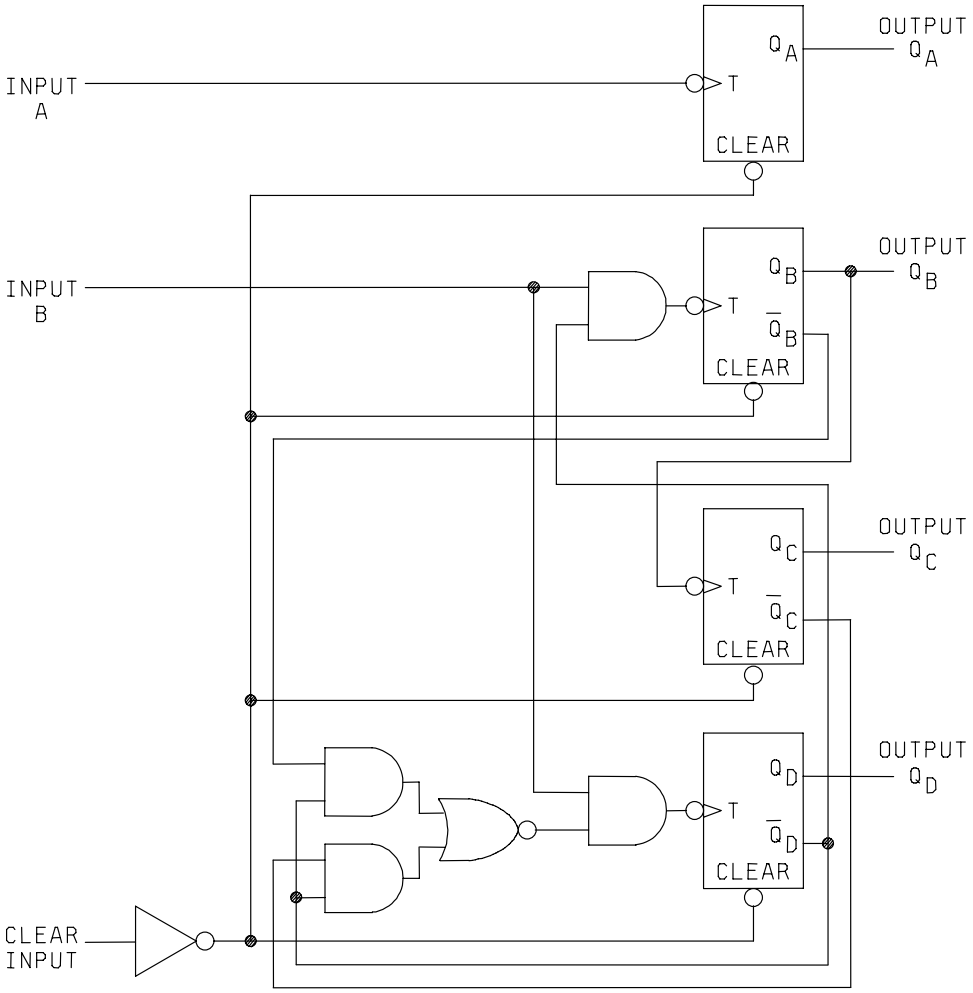


FIGURE 2. Logic diagrams (each counter).

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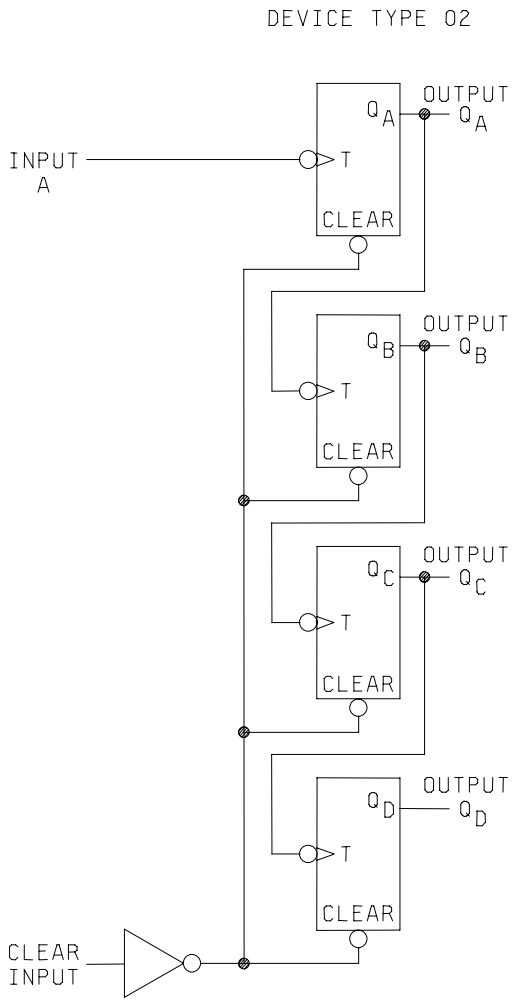


FIGURE 2. Logic diagrams (each counter) - Continued.

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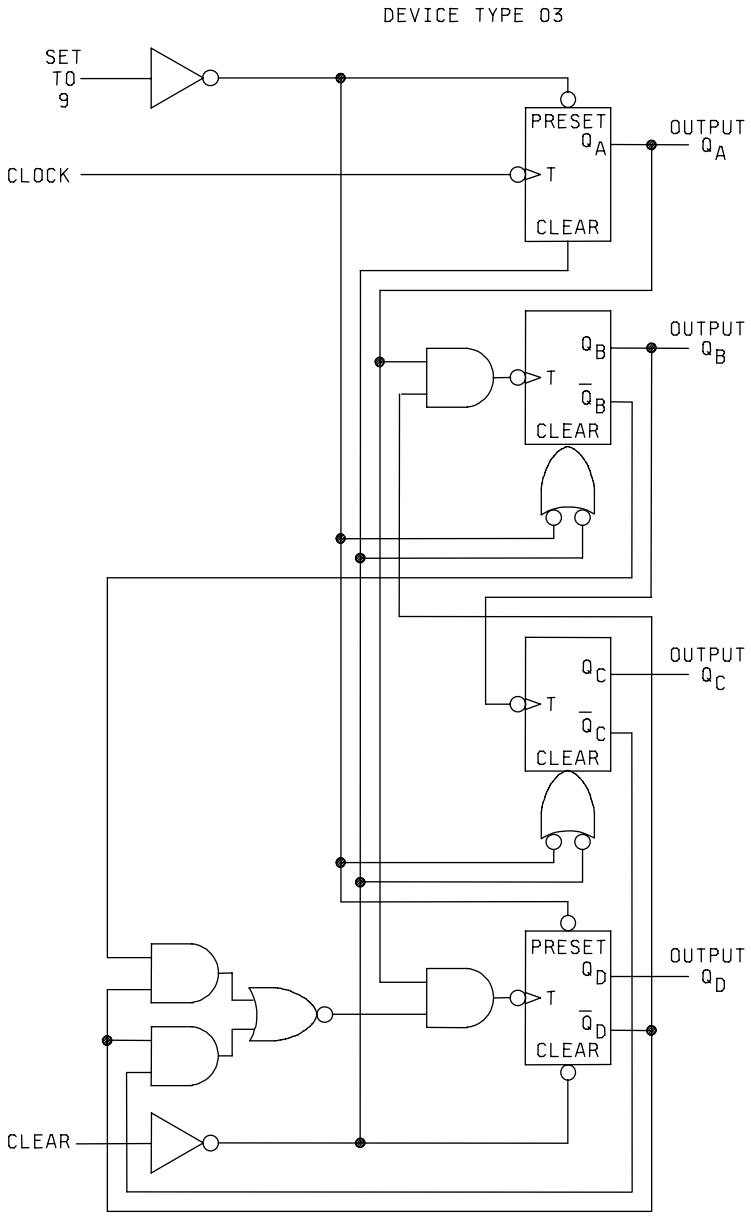


FIGURE 2. Logic diagrams (each counter) - Continued.

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Device types 01

EACH COUNTER

BCD COUNT SEQUENCE
(See Note A)

COUNT	OUTPUT			
	QD	QC	QB	QA
0	L	L	L	L
1	L	L	L	H
2	L	L	H	L
3	L	L	H	H
4	L	H	L	L
5	L	H	L	H
6	L	H	H	L
7	L	H	H	H
8	H	L	L	L
9	H	L	L	H

BI-QUINARY (5-2)
(See Note B)

COUNT	OUTPUT			
	QA	QD	QC	QB
0	L	L	L	L
1	L	L	L	H
2	L	L	H	L
3	L	L	H	H
4	L	H	L	L
5	H	L	L	L
6	H	L	L	H
7	H	L	H	L
8	H	L	H	H
9	H	H	L	L

NOTES:
A. Output QA is connected to input B for BCD count.
B. Output QD is connected to input A for bi-quinary count.

Device type 02

EACH COUNTER
COUNT SEQUENCE

COUNT	OUTPUT			
	QD	QC	QB	QA
0	L	L	L	L
1	L	L	L	H
2	L	L	H	L
3	L	L	H	H
4	L	H	L	L
5	L	H	L	H
6	L	H	H	L
7	L	H	H	H
8	H	L	L	L
9	H	L	L	H
10	H	L	H	L
11	H	L	H	H
12	H	H	L	L
13	H	H	L	H
14	H	H	H	L
15	H	H	H	H

FIGURE 3. Truth tables.

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Device type 03

BCD COUNT SEQUENCE
(EACH COUNTER)

COUNT	OUTPUT			
	QD	QC	QB	QA
0	L	L	L	L
1	L	L	L	H
2	L	L	H	L
3	L	L	H	H
4	L	H	L	L
5	L	H	L	H
6	L	H	H	L
7	L	H	H	H
8	H	L	L	L
9	H	L	L	H

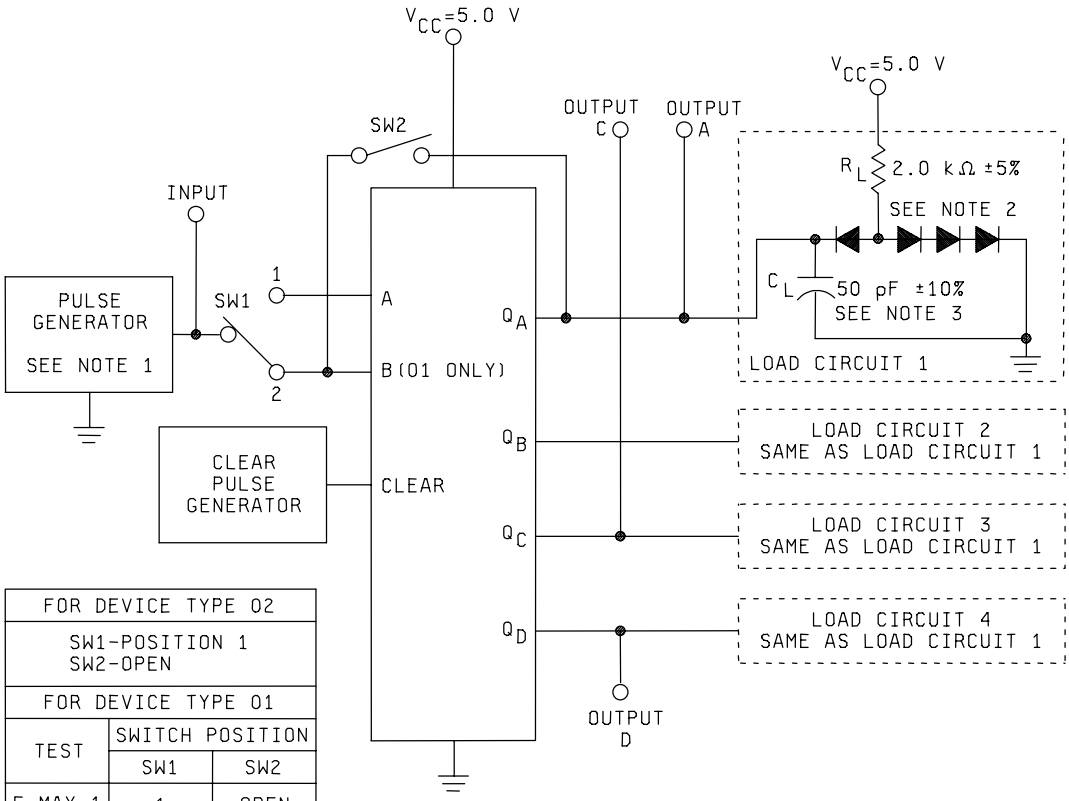
CLEAR/SET TO 9
FUNCTION TABLE
(EACH COUNTER)

INPUTS		OUTPUTS			
CLEAR	SET TO 9	QA	QB	QC	QD
H	L	L	L	L	L
L	H	H	L	L	H
L	L	COUNT			

H = high level, L = low level

FIGURE 3. Truth tables - Continued.

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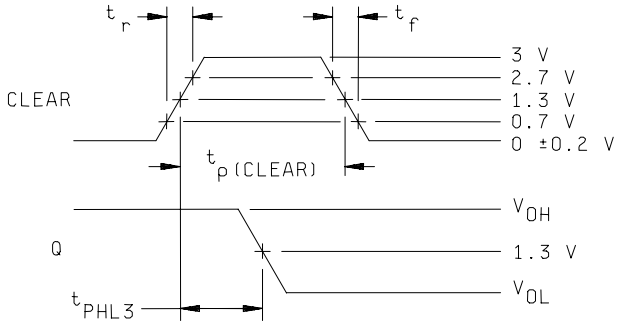


FOR DEVICE TYPE 02		
SW1-POSITION 1 SW2-OPEN		
FOR DEVICE TYPE 01		
TEST	SWITCH POSITION	
	SW1	SW2
F MAX 1	1	OPEN
F MAX 2	2	OPEN
A TO Q_C	1	CLOSED
B TO Q_D	2	OPEN

TEST CIRCUIT

NOTES:

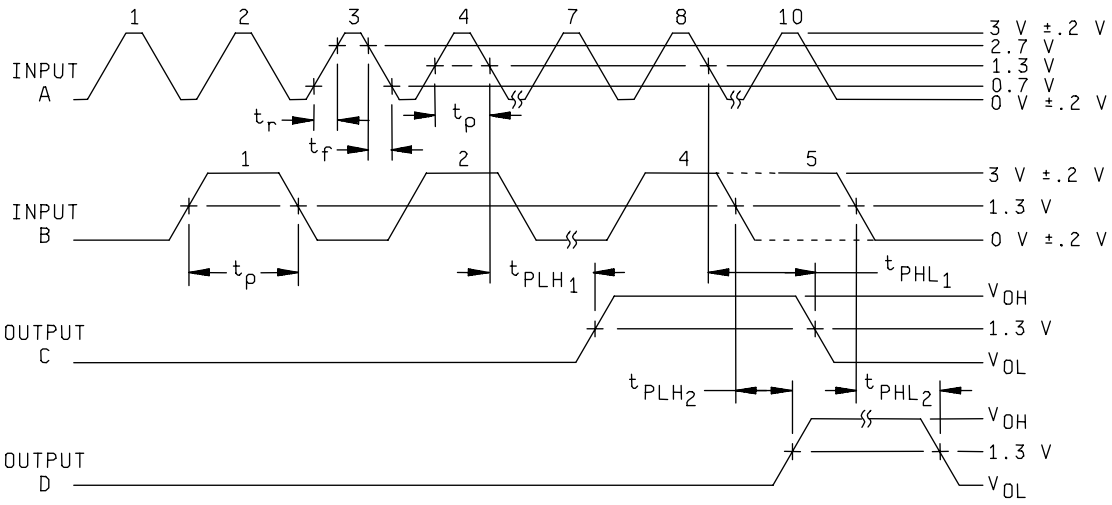
1. The pulse generator has the following characteristics: V_{gen} = 3 V, $t_r \leq 15$ ns, $t_f \leq 6$ ns, t_p = .5 µs, PRR ≤ 1 MHz, Zout ≈ 50, $t_{p(clear)} \geq 20$ ns, $t_{p(clear)} \geq 35$ ns for device 02.
2. All diodes are 1N3064 or equivalent.
3. C_L includes probe and jig capacitance.
4. Voltage values are with respect to ground teminal.
5. F maximum: $t_r = t_f \leq 6$ ns.



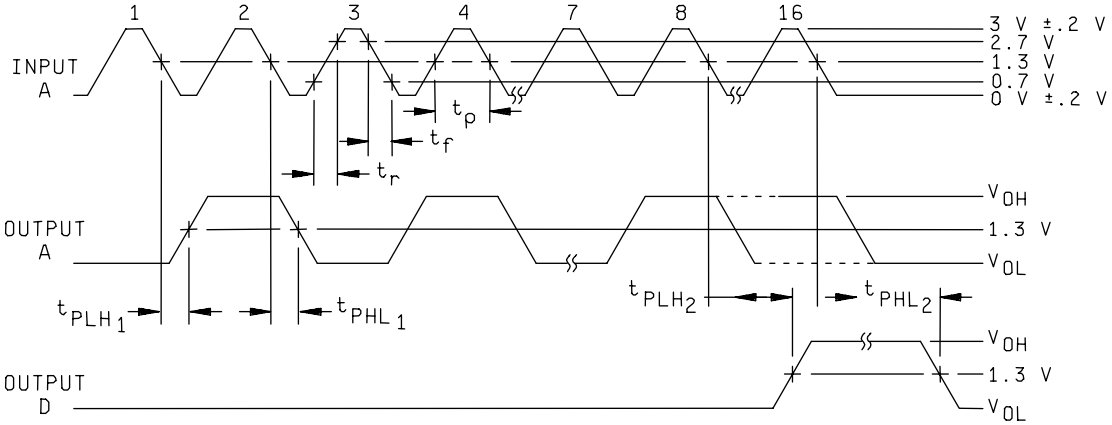
CLEAR VOLTAGE WAVEFORMS TYPES 01 AND 02

FIGURE 4. Switching time test circuit and waveforms for device types 01 and 02.

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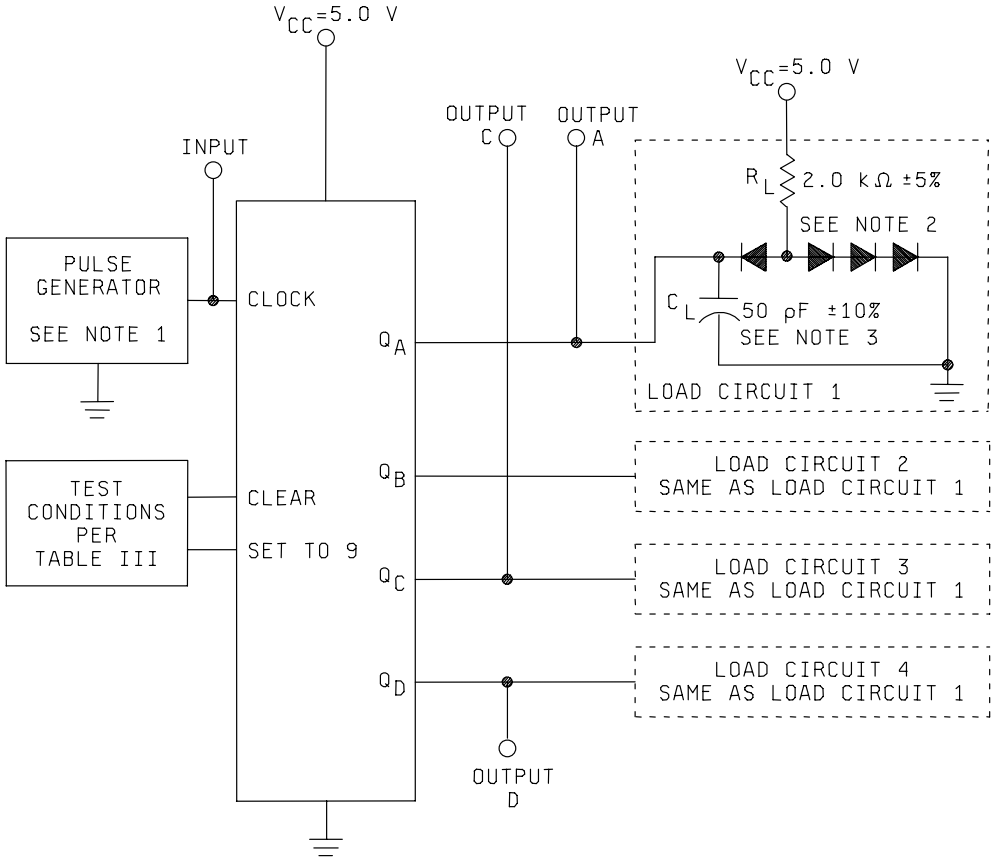
COUNT VOLTAGE WAVEFORMS, TYPE 01



COUNT VOLTAGE WAVEFORMS, TYPE 02

FIGURE 4. Switching time test circuit and waveforms for device types 01 and 02 - Continued.

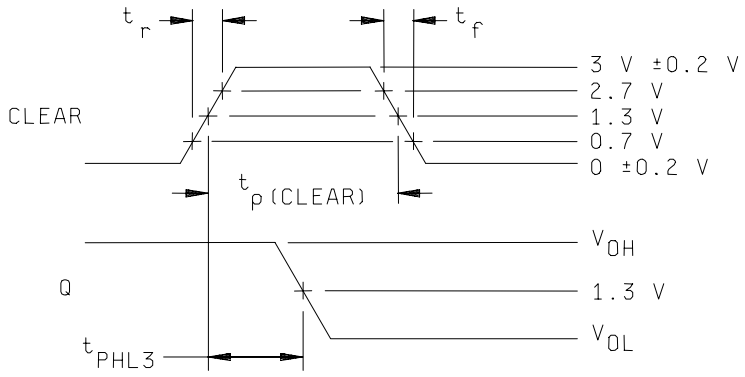
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TEST CIRCUIT

NOTES:

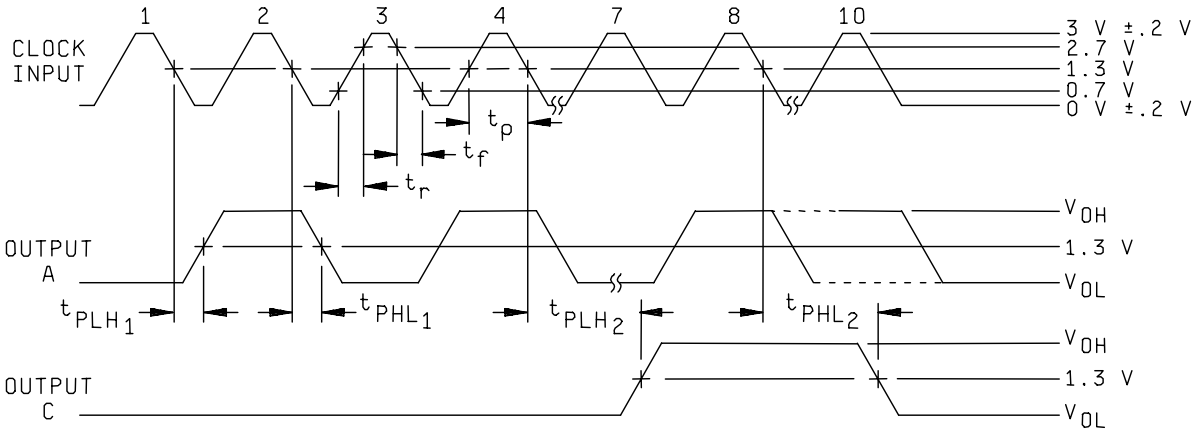
1. The pulse generator has the following characteristics: V_{gen} = 3 V, $t_r \leq 15$ ns, $t_f \leq 6$ ns, t_p = .5 µs, PRR ≤ 1 MHz, Zout ≈ 50, $t_{p(clear)} \geq 20$ ns.
2. All diodes are 1N3064 or equivalent.
3. C_L includes probe and jig capacitance.
4. Voltage values are with respect to ground teminal.
5. F maximum: $t_r = t_f \leq 6$ ns.



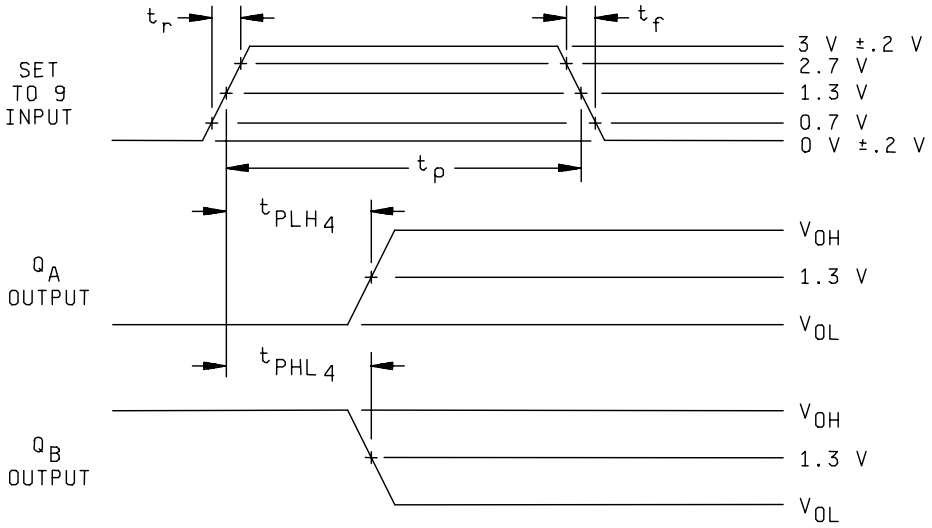
CLEAR VOLTAGE WAVEFORMS

FIGURE 5. Switching time test circuit and waveforms for device types 03.

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COUNT VOLTAGE WAVEFORMS



SET-TO-NINE VOLTAGE WAVEFORMS

FIGURE 5. Switching time test circuit and waveforms for device types 03 - Continued.

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TABLE III. Group A inspection for device type 01.

Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F / Case 2 / Test no.	1 / 2 / 1A	2 / 3 / 1CLR	3 / 4 / 1QA	4 / 5 / 1B	5 / 7 / 1QB	6 / 8 / 1QC	7 / 9 / 1QD	8 / 10 / GND	9 / 12 / 2QD	10 / 13 / 2QC	11 / 14 / 2QB	12 / 15 / 2B	13 / 17 / 2QA	14 / 18 / 2CLR	15 / 19 / 2A	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1	V_{OH}	3006	1	1/	2/	-0.4 mA					GND								4.5 V	1QA	2.5		V
Tc = 25°C		"	2		"		1/	-0.4 mA			"								"	1QB	"		"
		"	3		"		3/		-0.4 mA		"								"	1QC	"		"
		"	4		"		4/			-0.4 mA	"								"	1QD	"		"
		"	5								"					-0.4 mA	2/	1/	"	2QA	"		"
		"	6								"			-0.4 mA	1/		"		"	2QB	"		"
		"	7								"		-0.4 mA		3/		"		"	2QC	"		"
		"	8								"	-0.4 mA			4/		"		"	2QD	"		"
	V_{OL}	3007	9		2.0 V	5/					"								"	1QA		0.4	"
		"	10		"			4 mA			"								"	1QB		"	"
		"	11		"				4 mA		"								"	1QC		"	"
		"	12		"					4 mA	"								"	1QD		"	"
		"	13								"					5/	2.0 V		"	2QA		"	"
		"	14								"			4 mA			"		"	2QB		"	"
		"	15								"		4 mA				"		"	2QC		"	"
		"	16								"	4 mA					"		"	2QD		"	"
	V_{IC}		17	-18 mA							"								"	1A		-1.5	"
			18		-18 mA						"								"	1CLR		"	"
			19				-18 mA				"								"	1B		"	"
			20								"				-18 mA				"	2B		"	"
			21								"						-18 mA		"	2CLR		"	"
			22								"							-18 mA	"	2A		"	"
	I_{IL1}	3009	23		0.4 V						"								5.5 V	1CLR	6/	6/	6/
	I_{IL2}	"	24	0.4 V	GND						"								"	1A	"	"	"
	I_{IL3}	"	25		GND		0.4 V				"								"	1B	"	"	"
	I_{IL1}	"	26								"						0.4 V		"	2CLR	"	"	"
	I_{IL2}	"	27								"						GND	0.4 V	"	2A	"	"	"
	I_{IL3}	"	28								"				0.4 V		GND		"	2B	"	"	"
	I_{IH1}	3010	29		2.7 V						"								"	1CLR		20	µA
	I_{IH2}	"	30	2.7 V							"								"	1A		100	"
	I_{IH3}	"	31				2.7 V				"								"	1B		200	"
	I_{IH1}	"	32								"						2.7 V		"	2CLR		20	"
	I_{IH2}	"	33								"							2.7 V	"	2A		100	"
	I_{IH3}	"	34								"				2.7 V				"	2B		200	"
	I_{IH4}	"	35		5.5 V						"								"	1CLR		100	"
	I_{IH5}	"	36	5.5 V							"								"	1A		200	"
	I_{IH6}	"	37				5.5 V				"								"	1B		400	"
	I_{IH4}	"	38								"						5.5 V		"	2CLR		100	"
	I_{IH5}	"	39								"							5.5 V	"	2A		200	"
	I_{IH6}	"	40								"				5.5 V				"	2B		400	"
	I_{OS}	3011	41	1/	2/	GND					"								"	1QA	-15	-100	mA
		"	42		"		1/	GND			"								"	1QB	"	"	"
		"	43		"		3/		GND		"								"	1QC	"	"	"
		"	44		"		4/			GND	"								"	1QD	"	"	"
		"	45								"					GND	2/	1/	"	2QA	"	"	"
		"	46								"			GND	1/		"		"	2QB	"	"	"
		"	47								"		GND		3/		"		"	2QC	"	"	"
		"	48								"	GND			4/		"		"	2QD	"	"	"
	I_{CC}	3005	49	GND	2/	GND					"				GND		"	GND	"	V_{CC}		26	"
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						

See footnotes at end of device type 01.

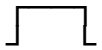
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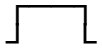
TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Case 2 Test no.	1 2 1A	2 3 1CLR	3 4 1QA	4 5 1B	5 7 1QB	6 8 1QC	7 9 1QD	8 10 GND	9 12 2QD	10 13 2QC	11 14 2QB	12 15 2B	13 17 2QA	14 18 2CLR	15 19 2A	16 20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
7 Tc = 25°C 7/	Truth table tests	3014	50 8/	A	A	L	A	L	L	L	GND	L	L	L	A	L	A	A	5.0 V		9/		
		"	51	A	"	"	B	"	"	"	"	"	"	"	B	"	"	A	"				
		"	52	B	"	"	B	"	"	"	"	"	"	"	B	"	"	B	"				
		"	53	"	"	"	A	"	"	"	"	"	"	"	A	"	"	"	"				
		"	54	"	B	"	A	"	"	"	"	"	"	"	A	"	B	"	"				
		"	55	"	"	"	B	H	"	"	"	"	"	H	B	"	"	"	"				
		"	56	"	"	"	A	H	"	"	"	"	"	H	A	"	"	"	"				
		"	57	"	"	"	B	L	H	"	"	"	H	L	B	"	"	"	"				
		"	58	"	"	"	A	L	"	"	"	"	"	L	A	"	"	"	"				
		"	59	"	"	"	B	H	"	"	"	"	"	H	B	"	"	"	"				
		"	60	"	"	"	A	H	"	"	"	"	"	H	A	"	"	"	"				
		"	61	"	"	"	B	L	L	H	"	H	L	L	B	"	"	"	"				
		"	62	"	"	"	A	"	"	H	"	H	"	"	A	"	"	"	"				
		"	63	"	"	"	B	"	"	L	"	L	"	"	B	"	"	"	"				
		"	64	A	"	"	"	"	"	"	"	"	"	"	"	"	"	A	"				
		"	65	B	"	H	"	"	"	"	"	"	"	"	"	H	"	B	"				
8	Same tests, terminal conditions, and limits as for subgroup 7 tests at T_C = +125°C and T_C = -55°C.																						
9 Tc = 25°C	F_{MAX1}	3003	66 10/	IN	GND	OUT					GND								5.0 V	1A to 1QA	25		MHz
	F_{MAX1}	Fig. 4	67 10/								"					OUT	GND	IN	"	2A to 2QA	25		"
	F_{MAX2}	"	68 10/		GND		IN			OUT	"								"	1B to 1QD	12.5		"
	F_{MAX2}	"	69 10/								"	OUT			IN		GND		"	2B to 2QD	12.5		"
	t_{PLH1}	"	70	IN	GND				OUT		"								"	1A to 1QC	2	65	ns
	t_{PLH1}	"	71								"		OUT				GND	IN	"	2A to 2QC	"	"	"
	t_{PHL1}	"	72	IN	GND				OUT		"								"	1A to 1QC	"	"	"
	t_{PHL1}	"	73								"		OUT				GND	IN	"	2A to 2QC	"	"	"
	t_{PLH2}	"	74		GND		IN			OUT	"								"	1B to 1QD	"	26	"
	t_{PLH2}	"	75								"	OUT			IN		GND		"	2B to 2QD	"	"	"
	t_{PHL2}	"	76		GND		IN			OUT	"								"	1B to 1QD	"	"	"
	t_{PHL2}	"	77								"	OUT			IN		GND		"	2B to 2QD	"	"	"
	t_{PHL3}	"	78	11/	IN	OUT					"								"	1CLR to 1QA	"	44	"
		"	79		"		11/	OUT			"								"	1CLR to 1QB	"	"	"
		"	80		"		"		OUT		"								"	1CLR to 1QC	"	"	"
		"	81		"		"			OUT	"								"	1CLR to 1QD	"	"	"
		"	82								"					OUT	IN	11/	"	2CLR to 2QA	"	"	"
		"	83								"			OUT	11/		"	"	"	2CLR to 2QB	"	"	"
		"	84								"		OUT		"		"	"	"	2CLR to 2QC	"	"	"
		"	85								"	OUT			"		"	"	"	2CLR to 2QD	"	"	"
10 Tc = 125°C	F_{MAX1}	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C.																			25		MHz
	F_{MAX2}																				12.5		MHz
	t_{PLH1}																				2	84	ns
	t_{PHL1}																				"	84	"
	t_{PLH2}																				"	34	"
	t_{PHL2}																				"	34	"
	t_{PHL3}																				"	56	"
11	Same tests, terminal conditions and limits as for subgroup 10, except T_C = -55°C.																						

See footnotes on the next page.

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1/ Apply  - - 2.0 V min./5.5 V max. prior to test after clear pulse.
0.0 ±0.2 V

2/ Apply  - - 2.0 V min./5.5 V max. pulse prior to test.
0.0 ±0.2 V

3/ Apply 2 pulses prior to test after clear pulse (see 1/).

4/ Apply 4 pulses prior to test after clear pulse (see 1/).

5/ For tests 9 and 13, I_{IL3} maximum value pulse 4 mA shall be applied to output QA.

6/ I_{IL} limits shall be as follows:

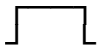
Symbol	Min/Max limits (mA)		
	Circuit		
	A	E	B
I_{IL1}	-.15/-.38	-.12/-.36	-.16/-.40
I_{IL2}	-.35/-1.6	-1.0/-2.4	-.35/-1.6
I_{IL3}	-.60/-2.4	-1.3/-3.2	-.60/-2.4

7/ Only a summary of attributes data is required.

8/ A = 2.4 V min. and B = 0.4 V max.

9/ Output voltages shall be either:
a. H ≥ 2.5 V and L ≤ 0.4 V when using a high speed double comparator, or
b. H ≥ 1.5 V and L ≤ 1.5 V when using a high speed single comparator.

10/ F_{MAX1} and F_{MAX2} minimum limits specified are the frequency of the input pulse.
The output pulse shall be one half the input frequency when measuring QA.
The output shall be one fifth of the input frequency when measuring QD.

11/ Apply sufficient  - - 2.0 V min./5.5 V max. pulses to set output high prior to test.
0.0 ±0.2 V

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TABLE III. Group A inspection for device type 02.

Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Case 2	2	3	4	6	8	9	10	12	13	14	16	18	19	20				
			Test no.	1A	1CLR	1QA	1QB	1QC	1QD	GND	2QD	2QC	2QB	2QA	2CLR	2A	V_{CC}		Min	Max	
1 Tc = 25°C	V_{OL}	3007	1		2.0 V	4 mA				GND							4.5 V	1QA		0.4	V
		"	2		"		4 mA			"							"	1QB		"	"
		"	3		"			4 mA		"							"	1QC		"	"
		"	4		"				4 mA	"							"	1QD		"	"
		"	5							"				4 mA	2.0 V		"	2QA		"	"
		"	6							"			4 mA		"		"	2QB		"	"
		"	7							"		4 mA			"		"	2QC		"	"
		"	8							"	4 mA				"		"	2QD		"	"
	V_{OH}	3006	9	1/	2/	-400 µA				"							"	1QA	2.5		"
		"	10	"	"		-400 µA			"							"	1QB	"		"
		"	11	"	"			-400 µA		"							"	1QC	"		"
		"	12	"	"				-400 µA	"							"	1QD	"		"
		"	13							"				-400 µA	2/	1/	"	2QA	"		"
		"	14							"			-400 µA		"	"	"	2QB	"		"
		"	15							"		-400 µA			"	"	"	2QC	"		"
		"	16							"	-400 µA				"	"	"	2QD	"		"
	V_{IC}		17	-18 mA						"							"	1A		-1.5	"
			18		-18 mA					"							"	1CLR		"	"
			19							"						-18 mA	"	2A		"	"
			20							"					-18 mA		"	2CLR		"	"
	I_{IL1}	3009	21		0.4 V					"							5.5 V	1CLR	3/	3/	3/
	I_{IL1}	"	22							"					0.4 V		"	2CLR	"	"	"
	I_{IL2}	"	23	0.4 V	GND					"							"	1A	"	"	"
	I_{IL2}	"	24							"					GND	0.4 V	"	2A	"	"	"
	I_{IH1}	3010	25		2.7 V					"							"	1CLR		20	µA
	I_{IH1}	"	26							"					2.7 V		"	2CLR		20	"
	I_{IH2}	"	27	2.7 V						"							"	1A		100	"
	I_{IH2}	"	28							"						2.7 V	"	2A		100	"
	I_{IH4}	"	29		5.5 V					"							"	1CLR		100	"
	I_{IH4}	"	30							"					5.5 V		"	2CLR		100	"
	I_{IH5}	"	31	5.5 V						"							"	1A		400	"
	I_{IH5}	"	32							"						5.5 V	"	2A		400	"
	I_{OS}	3011	33	1/	2/	GND				"							"	1QA	-15	-100	mA
		"	34	"	"		GND			"							"	1QB	"	"	"
		"	35	"	"			GND		"							"	1QC	"	"	"
		"	36	"	"				GND	"							"	1QD	"	"	"
		"	37							"				GND	2/	1/	"	2QA	"	"	"
		"	38							"			GND		"	"	"	2QB	"	"	"
		"	39							"		GND			"	"	"	2QC	"	"	"
		"	40							"	GND				"	"	"	2QD	"	"	"
	I_{CC}	3005	41	GND	2/					"					"	GND	"	V_{CC}		26	"
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open).

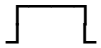
Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Case 2	2	3	4	6	8	9	10	12	13	14	16	18	19	20				
			Test no.	1A	1CLR	1QA	1QB	1QC	1QD	GND	2QD	2QC	2QB	2QA	2CLR	2A	V_{CC}		Min	Max	
7 4/ Tc = 25°C	Truth table tests	3014	42	A	A	L	L	L	L	GND	L	L	L	L	A	A	5.0 V	5/			
		"	43	B	A	"	"	"	"	"	"	"	"	"	A	B	"				
		"	44	B	B	"	"	"	"	"	"	"	"	"	B	B	"				
		"	45	A	"	"	"	"	"	"	"	"	"	"	"	A	"				
		"	46	B	"	H	"	"	"	"	"	"	"	H	"	B	"				
		"	47	A	"	H	"	"	"	"	"	"	"	H	"	A	"				
		"	48	B	"	L	H	"	"	"	"	"	H	L	"	B	"				
		"	49	A	"	L	"	"	"	"	"	"	"	L	"	A	"				
		"	50	B	"	H	"	"	"	"	"	"	"	H	"	B	"				
		"	51	A	"	H	"	"	"	"	"	"	"	H	"	A	"				
		"	52	B	"	L	L	H	"	"	"	H	L	L	"	B	"				
		"	53	A	"	L	"	"	"	"	"	"	"	L	"	A	"				
		"	54	B	"	H	"	"	"	"	"	"	"	H	"	B	"				
		"	55	A	"	H	"	"	"	"	"	"	"	H	"	A	"				
		"	56	B	"	L	H	"	"	"	"	"	H	L	"	B	"				
		"	57	A	"	L	"	"	"	"	"	"	"	L	"	A	"				
		"	58	B	"	H	"	"	"	"	"	"	"	H	"	B	"				
		"	59	A	"	H	"	"	"	"	"	"	"	H	"	A	"				
		"	60	B	"	L	L	L	H	"	H	L	L	L	"	B	"				
		"	61	A	"	L	"	"	"	"	"	"	"	L	"	A	"				
		"	62	B	"	H	"	"	"	"	"	"	"	H	"	B	"				
		"	63	A	"	H	"	"	"	"	"	"	"	H	"	A	"				
		"	64	B	"	L	H	"	"	"	"	"	H	L	"	B	"				
		"	65	A	"	L	"	"	"	"	"	"	"	L	"	A	"				
		"	66	B	"	H	"	"	"	"	"	"	"	H	"	B	"				
		"	67	A	"	H	"	"	"	"	"	"	"	H	"	A	"				
		"	68	B	"	L	L	H	"	"	"	H	L	L	"	B	"				
		"	69	A	"	L	"	"	"	"	"	"	"	L	"	A	"				
		"	70	B	"	H	"	"	"	"	"	"	"	H	"	B	"				
		"	71	A	"	H	"	"	"	"	"	"	"	H	"	A	"				
		"	72	B	"	L	H	"	"	"	"	"	H	L	"	B	"				
		"	73	A	"	L	"	"	"	"	"	"	"	L	"	A	"				
		"	74	B	"	H	"	"	"	"	"	"	"	H	"	B	"				
		"	75	B	A	L	L	L	L	"	L	L	L	L	A	B	"				
8	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and T_C = -55°C.																				
9	F_{MAX1}	3003	76 6/	IN	GND	OUT				GND							5.0 V	1A to 1QA	25		MHz
Tc = 25°C	F_{MAX2}	"	77 6/							"				OUT	GND	IN	"	2A to 2QA	25		MHz
	t_{PLH1}	"	78	IN	GND	OUT				"							"	1A to 1QA	2	25	ns
	t_{PLH1}	"	79							"				OUT	GND	IN	"	2A to 2QA	"	"	"
	t_{PHL1}	"	80	IN	GND	OUT				"							"	1A to 1QA	"	"	"
	t_{PHL1}	"	81							"				OUT	GND	IN	"	2A to 2QA	"	"	"
	t_{PLH2}	"	82	IN	GND				OUT	"							"	1A to 1QD	"	70	"
	t_{PLH2}	"	83							"	OUT				GND	IN	"	2A to 2QD	"	"	"
	t_{PHL2}	"	84	IN	GND				OUT	"							"	1A to 1QD	"	"	"
	t_{PHL2}	"	85							"	OUT				GND	IN	"	2A to 2QD	"	"	"
	t_{PHL3}	"	86	7/	IN	OUT				"							"	1CLR to 1QA	"	44	"
		"	87	"	"		OUT			"							"	1CLR to 1QB	"	"	"
		"	88	"	"			OUT		"							"	1CLR to 1QC	"	"	"
		"	89	"	"				OUT	"							"	1CLR to 1QD	"	"	"
		"	90							"				OUT	IN	7/	"	2CLR to 2QA	"	"	"
		"	91							"			OUT		"	"	"	2CLR to 2QB	"	"	"
		"	92							"		OUT			"	"	"	2CLR to 2QC	"	"	"
		"	93							"	OUT				"	"	"	2CLR to 2QD	"	"	"

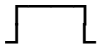
See footnotes at end of device types 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Case 2	2	3	4	6	8	9	10	12	13	14	16	18	19	20				
			Test no.	1A	1CLR	1QA	1QB	1QC	1QD	GND	2QD	2QC	2QB	2QA	2CLR	2A	V_{CC}		Min	Max	
10 T_C = 125°C	F_{MAX1}	Same tests and terminal conditions as for subgroup 9, except T_C = 125°C.																	25		MHz
	t_{PLH1}																		2	33	ns
	t_{PHL1}																		"	33	"
	t_{PLH2}																		"	93	"
	t_{PHL2}																		"	93	"
	t_{PHL3}																		"	56	"
11 T_C = -55°C		Same tests and terminal conditions as for subgroup 10, except T_C = -55°C.																			

1/ Apply  - - 2.0 V min./5.5 V max. to clear input and then to A input sufficient times prior to test to set the output high.
) ±0.2 V

2/ Apply  - - 2.0 V min./5.5 V max. pulse prior to test.
0.0 ±0.2 V

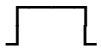
3/ I_{IL} limits shall be as follows:

Test	Min/Max limits (mA)				
	Circuits				
	A	E	B	D	F
I_{IL1}	-.15/-.38	-.12/-.36	-.16/-.40	-.12/-.36	-.12/-.36
I_{IL2}	-.35/-1.60	-1.0/-2.4	-.35/-1.60	-.35/-1.60	-.35/-1.60

4/ A = 2.4 V min. and B = 0.4 V max.

5/ Output voltages shall be either:
a. H ≥ 2.5 V and L ≤ 0.4 V max when using a high speed checker double comparator, or
b. H ≥ 1.5 V and L ≤ 1.5 V when using a high speed checker single comparator.

6/ F_{MAX1} minimum limit specified is the frequency of the input pulse.
The output pulse shall be one half the input frequency.

7/ Apply sufficient  - - 2.0 V min./5.5 V max. pulses to set output high prior to test.
0.0 ±0.2 V

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TABLE III. Group A inspection for device type 03.

Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Case 2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Test no.	1CLK	1CLR	1QA	1ST9	1QB	1QC	1QD	GND	2QD	2QC	2QB	2ST9	2QA	2CLR	2CLK	V_{CC}		Min	Max	
1 Tc = 25°C	V_{OH}	3006	1	1/	2/	-0.4 mA	0.7 V				GND								4.5 V	1QA	2.5		V
		"	2	3/	"		"	-0.4 mA			"								"	1QB	"		"
		"	3	4/	"		"		-0.4 mA		"								"	1QC	"		"
		"	4		0.7 V		2.0 V			-0.4 mA	"								"	1QD	"		"
		"	5								"				0.7 V	-0.4 mA	2/	1/	"	2QA	"		"
		"	6								"			-0.4 mA	"		"	3/	"	2QB	"		"
		"	7								"		-0.4 mA		"		"	4/	"	2QC	"		"
		"	8								"	-0.4 mA			2.0 V		0.7 V		"	2QD	"		"
	V_{OL}	3007	9		2.0 V	4 mA	0.7 V				"								"	1QA		0.4	"
		"	10		"		"	4 mA			"								"	1QB		"	"
		"	11		"		"		4 mA		"								"	1QC		"	"
		"	12		"		"			4 mA	"								"	1QD		"	"
		"	13								"				0.7 V	4 mA	2.0 V		"	2QA		"	"
		"	14								"			4 mA	"		"		"	2QB		"	"
		"	15								"		4 mA		"		"		"	2QC		"	"
		"	16								"	4 mA			"		"		"	2QD		"	"
	V_{IC}		17	-18 mA							"								"	1CLK		-1.5	"
			18		-18 mA						"								"	1CLR		"	"
			19				-18 mA				"								"	1ST9		"	"
			20								"				-18 mA				"	2ST9		"	"
			21								"						-18 mA		"	2CLR		"	"
			22								"							-18 mA	"	2CLK		"	"
	I_{IL1}	3009	23		0.4 V						"								5.5 V	1CLR	5/	5/	5/
	I_{IL2}	"	24	0.4 V	GND		GND				"								"	1CLK	"	"	"
	I_{IL1}	"	25				0.4 V				"								"	1ST9	"	"	"
	I_{IL1}	"	26								"						0.4 V		"	2CLR	"	"	"
	I_{IL2}	"	27								"				GND		GND	0.4 V	"	2CLK	"	"	"
	I_{IL1}	"	28								"				0.4 V				"	2ST9	"	"	"
	I_{IH1}	3010	29		2.7 V						"								"	1CLR		20	µA
	I_{IH2}	"	30	2.7 V							"								"	1CLK		100	"
	I_{IH1}	"	31				2.7 V				"								"	1ST9		20	"
	I_{IH1}	"	32								"						2.7 V		"	2CLR		20	"
	I_{IH2}	"	33								"							2.7 V	"	2CLK		100	"
	I_{IH1}	"	34								"				2.7 V				"	2ST9		20	"
	I_{IH4}	"	35		5.5 V						"								"	1CLR		100	"
	I_{IH5}	"	36	5.5 V							"								"	1CLK		200	"
	I_{IH4}	"	37				5.5 V				"								"	1ST9		100	"
	I_{IH4}	"	38								"						5.5 V		"	2CLR		100	"
	I_{IH5}	"	39								"							5.5 V	"	2CLK		200	"
	I_{IH4}	"	40								"				5.5 V				"	2ST9		100	"
	I_{OS}	3011	41	1/	2/	GND	GND				"								"	1QA	-15	-100	mA
		"	42	3/	"		"	GND			"								"	1QB	"	"	"
		"	43	4/	"		"		GND		"								"	1QC	"	"	"
		"	44		GND		5.5 V			GND	"								"	1QD	"	"	"
		"	45								"				GND	GND	2/	1/	"	2QA	"	"	"
		"	46								"			GND	"		"	3/	"	2QB	"	"	"
		"	47								"		GND		"		"	4/	"	2QC	"	"	"
		"	48								"	GND			5.5 V		GND		"	2QD	"	"	"
	I_{CC}	3005	49	GND	2/		GND				"				GND		2/	GND	"	V_{CC}		26	"
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						

See footnotes at end of device type 03.

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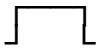
TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Case 2 Test no.	1 2 1CLK	2 3 1CLR	3 4 1QA	4 5 1ST9	5 7 1QB	6 8 1QC	7 9 1QD	8 10 GND	9 12 2QD	10 13 2QC	11 14 2QB	12 15 2ST9	13 17 2QA	14 18 2CLR	15 19 2CLK	16 20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
7 Tc = 25°C 6/	Truth table tests	3014	50 7/	A	A	L	B	L	L	L	GND	L	L	L	B	L	A	A	5.0 V		8/		
		"	51	B	A	"	"	"	"	"	"	"	"	"	"	"	A	B	"				
		"	52	A	B	"	"	"	"	"	"	"	"	"	"	"	B	A	"				
		"	53	B	"	H	"	"	"	"	"	"	"	"	"	H	"	B	"				
		"	54	A	"	H	"	"	"	"	"	"	"	"	"	H	"	A	"				
		"	55	B	"	L	"	H	"	"	"	"	"	H	"	L	"	B	"				
		"	56	A	"	L	"	"	"	"	"	"	"	"	"	L	"	A	"				
		"	57	B	"	H	"	"	"	"	"	"	"	"	"	H	"	B	"				
		"	58	A	"	H	"	"	"	"	"	"	"	"	"	H	"	A	"				
		"	59	B	"	L	"	L	H	"	"	"	H	L	"	L	"	B	"				
		"	60	A	"	L	"	"	"	"	"	"	"	"	"	L	"	A	"				
		"	61	B	"	H	"	"	"	"	"	"	"	"	"	H	"	B	"				
		"	62	A	"	H	"	"	"	"	"	"	"	"	"	H	"	A	"				
		"	63	B	"	L	"	H	"	"	"	"	"	H	"	L	"	B	"				
		"	64	A	"	L	"	"	"	"	"	"	"	"	"	L	"	A	"				
		"	65	B	"	H	"	"	"	"	"	"	"	"	"	H	"	B	"				
		"	66	A	"	H	"	"	"	"	"	"	"	"	"	H	"	A	"				
		"	67	B	"	L	"	L	L	H	"	H	L	L	"	L	"	B	"				
		"	68	A	"	L	"	"	"	"	"	"	"	"	"	L	"	A	"				
		"	69	B	"	H	"	"	"	"	"	"	"	"	"	H	"	B	"				
		"	70	A	A	L	"	"	"	L	"	L	"	"	"	L	A	A	"				
		"	71	"	B	L	"	"	"	L	"	L	"	"	"	L	B	"	"				
		"	72	"	B	H	A	"	"	H	"	H	"	"	A	H	B	"	"				
8	Same tests, terminal conditions, and limits as for subgroup 7 tests at T_C = +125°C and T_C = -55°C.																					
9 Tc = 25°C	F_{MAX1}	3003	73 9/	IN	GND	OUT	GND				GND								5.0 V	CLK to 1QA	25		MHz
	F_{MAX1}	Fig. 5	74 9/								"				GND	OUT	GND	IN	"	CLK to 2QA	25		MHz
	t_{PLH4}	"	75		2/	OUT	IN				"								"	ST9 to 1QA	2	44	ns
	t_{PLH4}	"	76								"				IN	OUT	2/		"	ST9 to 2QA	"	44	"
	t_{PLH1}	"	77	IN	GND	OUT	GND				"								"	CLK to 1QA	"	25	"
	t_{PLH1}	"	78								"				GND	OUT	GND	IN	"	CLK to 2QA	"	"	"
	t_{PHL1}	"	79	IN	GND	OUT	GND				"								"	CLK to 1QA	"	"	"
	t_{PHL1}	"	80								"				GND	OUT	GND	IN	"	CLK to 2QA	"	"	"
	t_{PLH2}	"	81	IN	GND		GND		OUT		"								"	CLK to 1QC	"	59	"
	t_{PLH2}	"	82								"		OUT		GND		GND	IN	"	CLK to 2QC	"	"	"
	t_{PHL2}	"	83	IN	GND		GND		OUT		"								"	CLK to 1QC	"	"	"
	t_{PHL2}	"	84								"		OUT		GND		GND	IN	"	CLK to 2QC	"	"	"
	t_{PHL3}	"	85	10/	IN	OUT	GND				"								"	CLR to 1QA	"	44	"
		"	86	"	"		"	OUT			"								"	CLR to 1QB	"	"	"
		"	87	"	"		"		OUT		"								"	CLR to 1QC	"	"	"
		"	88	"	"		"			OUT	"								"	CLR to 1QD	"	"	"
		"	89								"				GND	OUT	IN	10/	"	CLR to 2QA	"	"	"
		"	90								"			OUT	"		"	"	"	CLR to 2QB	"	"	"
		"	91								"		OUT		"		"	"	"	CLR to 2QC	"	"	"
		"	92								"	OUT			"		"	"	"	CLR to 2QD	"	"	"
	t_{PHL4}	"	93	10/	GND		IN	OUT			"								"	ST9 to 1QB	"	41	"
	t_{PHL4}	"	94								"			OUT	IN		GND	10/	"	ST9 to 2QB	"	41	"
10 Tc = 125°C	F_{MAX1}	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C.																			25		MHz
	t_{PLH4}																				2	57	ns
	t_{PLH1}																				"	33	"
	t_{PHL1}																				"	33	"
	t_{PLH2}																				"	77	"
	t_{PHL2}																				"	77	"
	t_{PHL3}																				"	56	"
	t_{PHL4}																				"	53	"
11	Same tests, terminal conditions and limits as for subgroup 10, except T_C = -55°C.																						

See footnotes on next page.

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1/ Apply  - - 2.0 V min./5.5 V max. prior to test after clear pulse.
0.0 ±0.2 V

2/ Apply  - - 2.0 V min./5.5 V max. pulse prior to test.
0.0 ±0.2 V

3/ Apply 2 pulses after clear pulse (see 1/).

4/ Apply 4 pulses after clear pulse (see 1/).

5/ I_{IL} limits shall be as follows:

Symbol	Min/Max limits (mA)		
	Circuit		
	A	E	B
I_{IL1} (CLR)	-.15/-.38	-.135/-.37	-.16/-.40
I_{IL2} (ST9)	-.16/-.40	-.135/-.37	-.16/-.40
I_{IL3}	-.35/-1.6	-1.0/-2.4	-.35/-1.6

6/ Only a summary of attributes data is required.

7/ A = 2.4 V min. and B = 0.4 V max.

8/ Output voltages shall be either:
a. H ≥ 2.5 V and L ≤ 0.4 V when using a high speed double comparator, or
b. H ≥ 1.5 V and L ≤ 1.5 V when using a high speed single comparator.

9/ F_{MAX1} minimum limits specified is the frequency of the input pulse. The output pulse shall be one half the input frequency.

10/ Apply sufficient  - - 2.0 V min./5.5 V max. pulses to set output high prior to test.
0.0 ±0.2 V

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5. PACKAGING

5.1 Packaging requirements. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Complete part number (see 1.2).

c. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

d. Requirements for certificate of compliance, if applicable.

e. Requirements for notification of change of product or process to contracting activity in addition to notification to the qualifying activity, if applicable.

f. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

g. Requirements for product assurance options.

h. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements should not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

I. Requirements for "JAN" marking.

6.3 Superseding information. The requirements of MIL-M-38510 have been superseded to take advantage of the available Qualified Manufacturer Listing (QML) system provided by MIL-PRF-38535. Previous references to MIL-M-38510 in this document have been replaced by appropriate references to MIL-PRF-38535. All technical requirements now consist of this specification and MIL-PRF-38535. The MIL-M-38510 specification sheet number and PIN have been retained to avoid adversely impacting existing government logistics systems and contractor's parts lists.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List QML-38535 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DSCC-VQ, 3990 E. Broad Street, Columbus, Ohio 43123-1199.

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6.5 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-PRF-38535, MIL-HDBK-1331, and as follows:

F_{MAX} ..... Maximum operating frequency
GND ..... Ground zero voltage potential
I_{IN} ..... Current flowing into an input terminal
V_{IC} ..... Input clamp voltage
V_{IN} ..... Voltage level at an input terminal

6.6 Logistic support. Lead materials and finishes (see 3.4) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish A (see 3.4). Longer length leads and lead forming should not affect the part number.

6.7 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information should not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-PRF-38535.

Military device type	Generic-industry type
01	54LS390
02	54LS393
03	54LS490

6.8 Manufacturers' designation. Manufacturers' circuits, which form a part of this specification, are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturer's designator.

Device type	Manufacturer				
	Texas Instruments	Signetics Corp.	Motorola Inc.	Fairchild Co.	National Semiconductor
	Circuit A	Circuit B	Circuit D	Circuit E	Circuit F
01	X	X		X	
02	X	X	X	X	X
03	X	X		X	

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6.9 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5962-1977)

Review activities:
Army - MI, SM
Navy - AS, CG, MC, SH, TD
Air Force - 03, 19, 99

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7, and send to preparing activity.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-M-38510/327B	2. DOCUMENT DATE *(YYYYMMDD)* 2003-09-15

3. DOCUMENT TITLE
MICROCIRCUITS, DIGITAL, BIPOLAR, LOW-POWER SCHOTTKY TTL, COUNTERS, MONOLITHIC SILICON

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) DSN *(If applicable)*	7. DATE SUBMITTED *(YYYYMMDD)*

8. PREPARING ACTIVITY

a. NAME Defense Supply Center, Columbus	b. TELEPHONE *(Include Area Code* (1) Commercial 614-692-0536 (2) DSN 850-0536
c. ADDRESS *(Include Zip Code)* DSCC-VA P. O. Box 3990 Columbus, Ohio 43216-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvoir, Virginia 22060-6221 Telephone (703)767-6888 DSN 427-6888

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